

Taking Picture of a Teapot



How to take a picture to capture the most information
about the teapot?



Most people will choose the top position



Why & How ?





Best position for a teapot snapshot

Why this position?

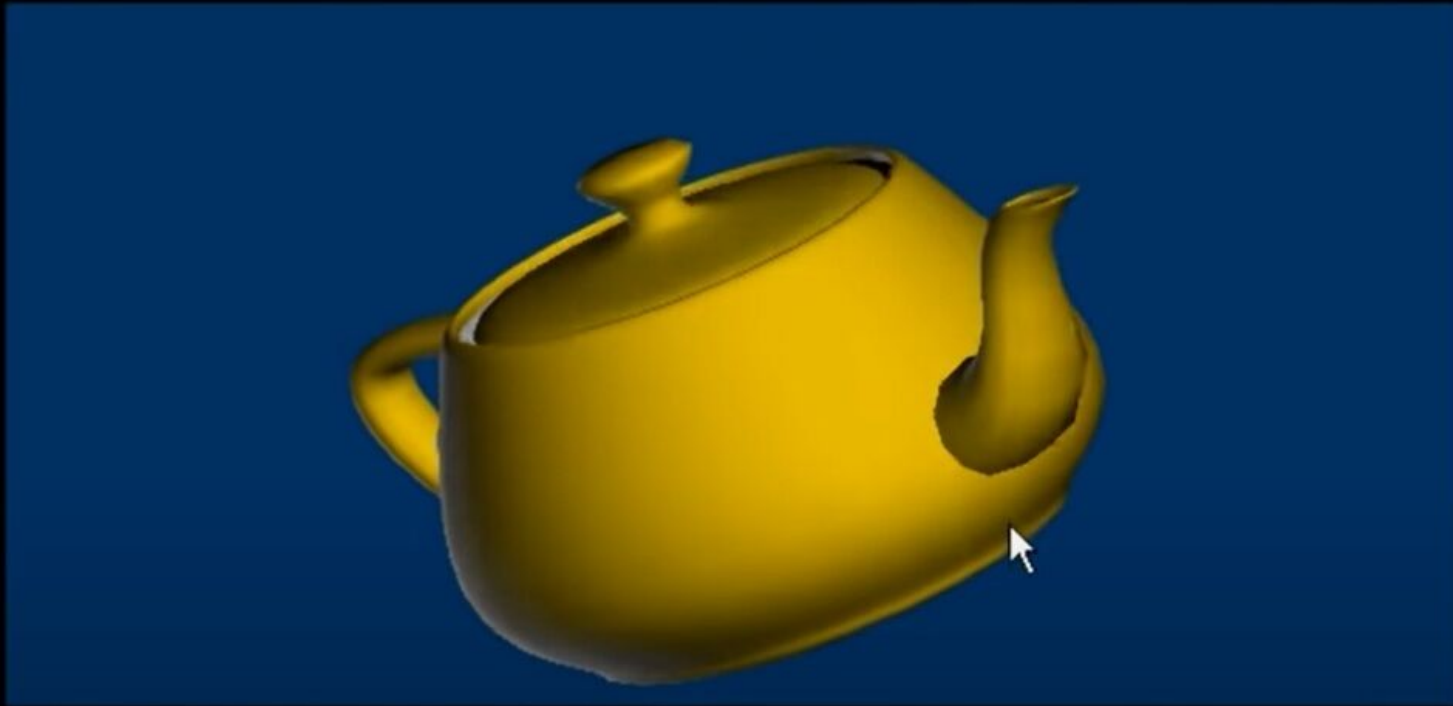
Because it provides the most visual information.

How do we find this position ?

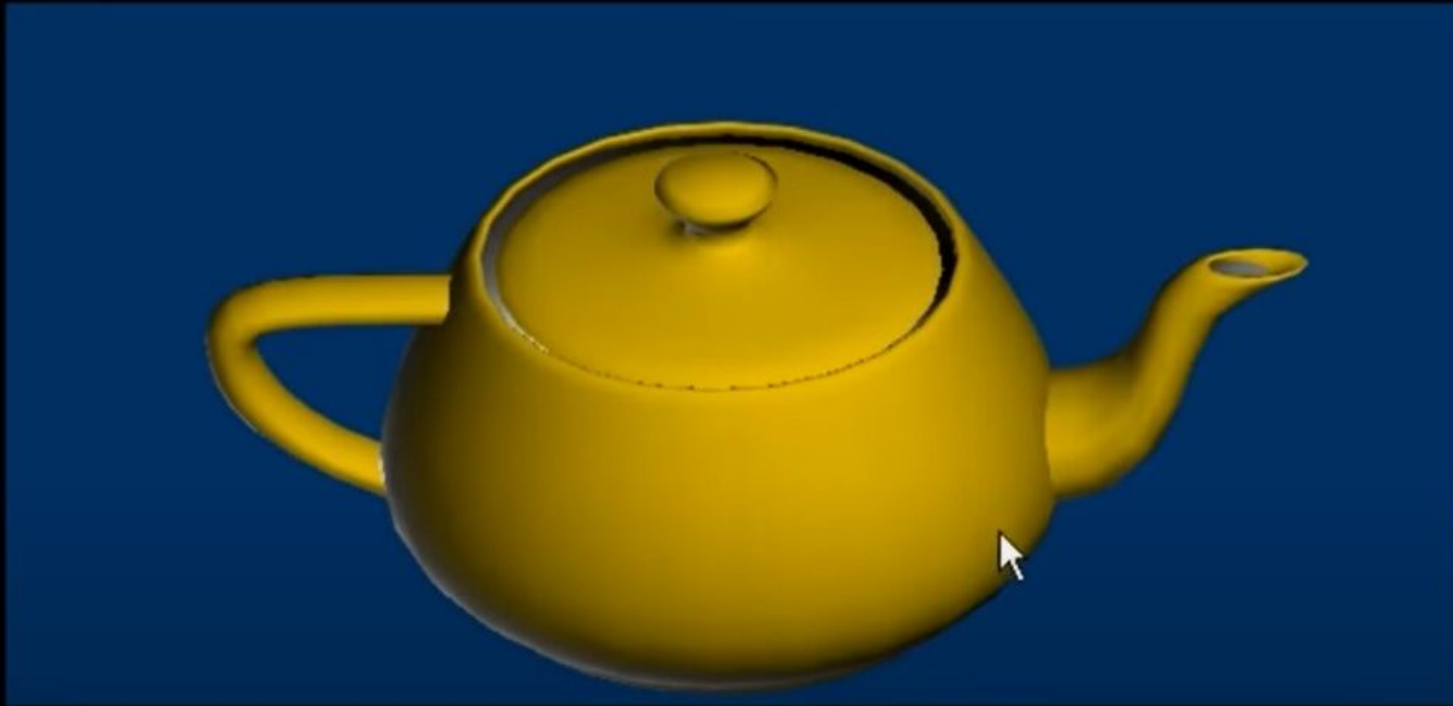
Rotate the teapot according to the PCA algorithm.

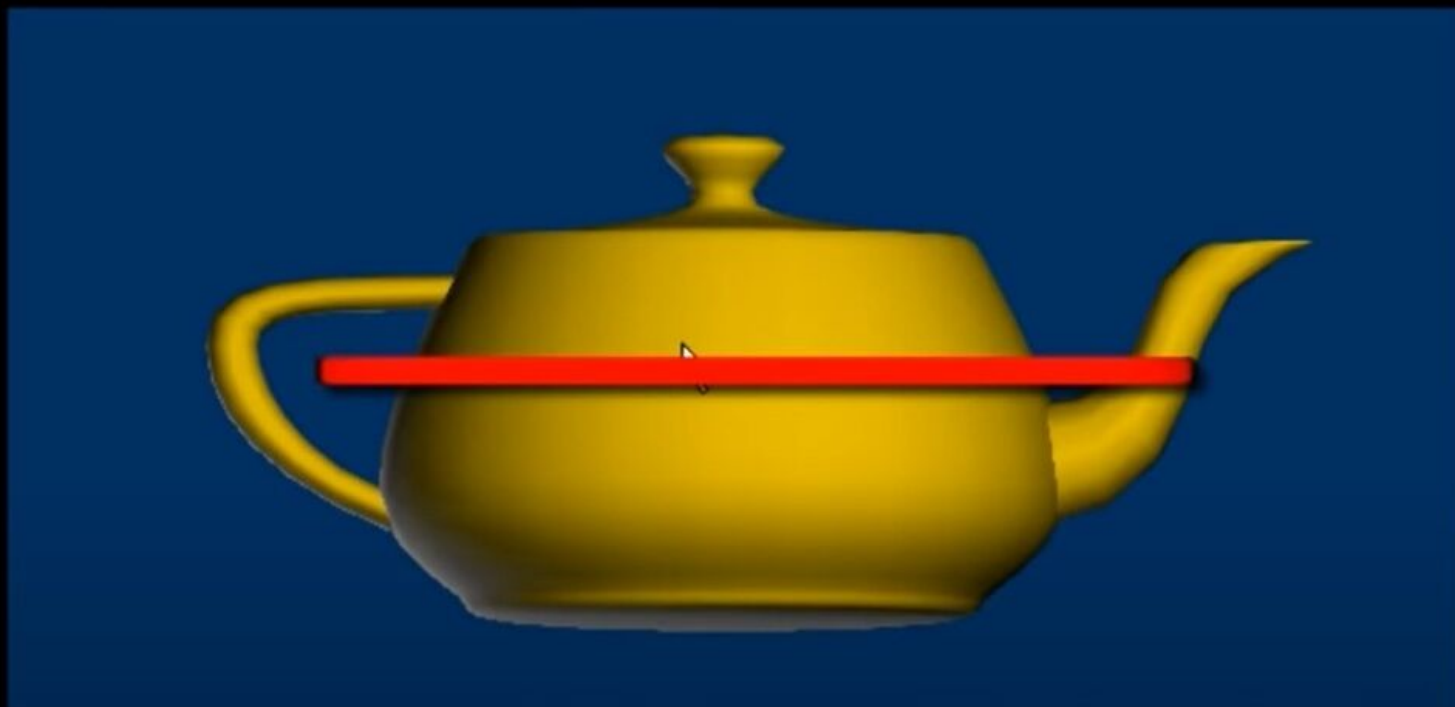


Finding the longest axis...

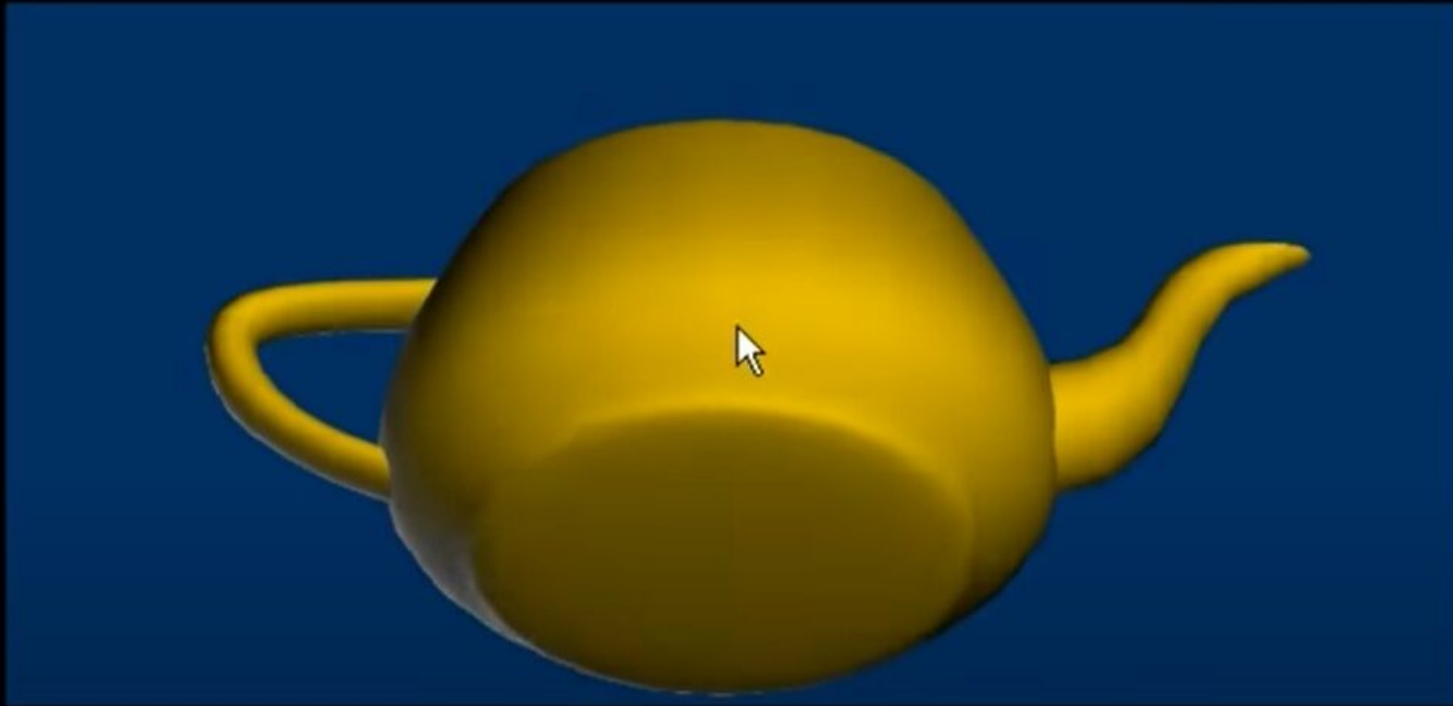


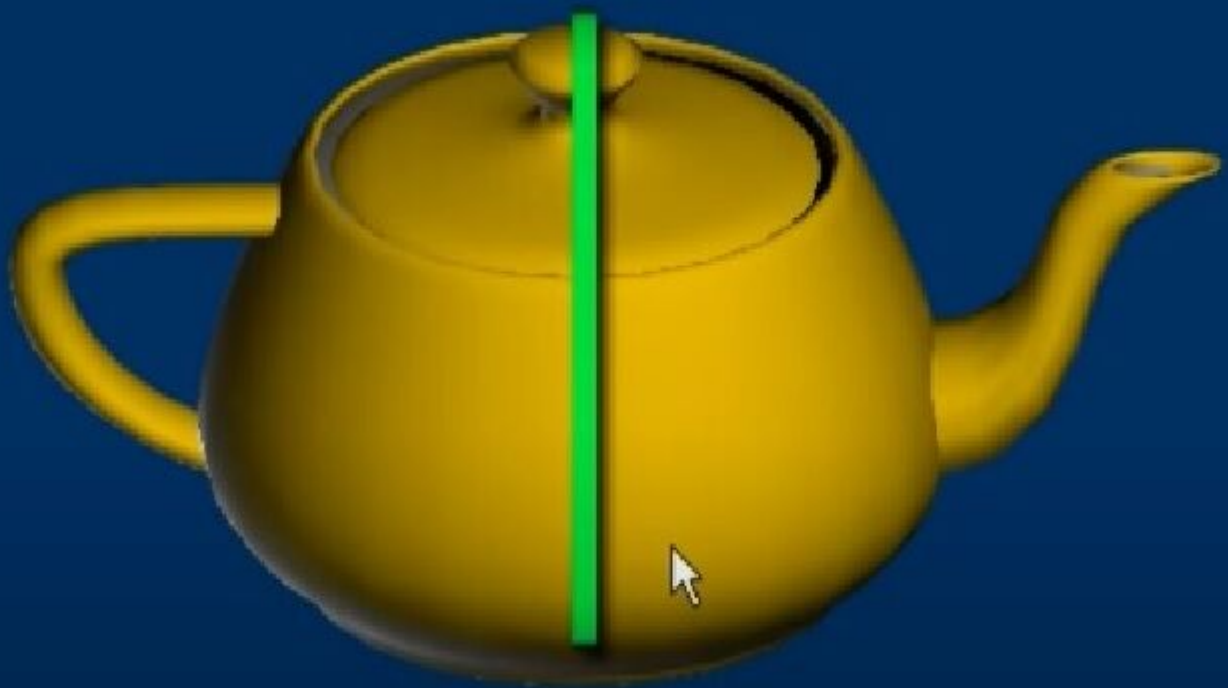
Finding the longest axis...

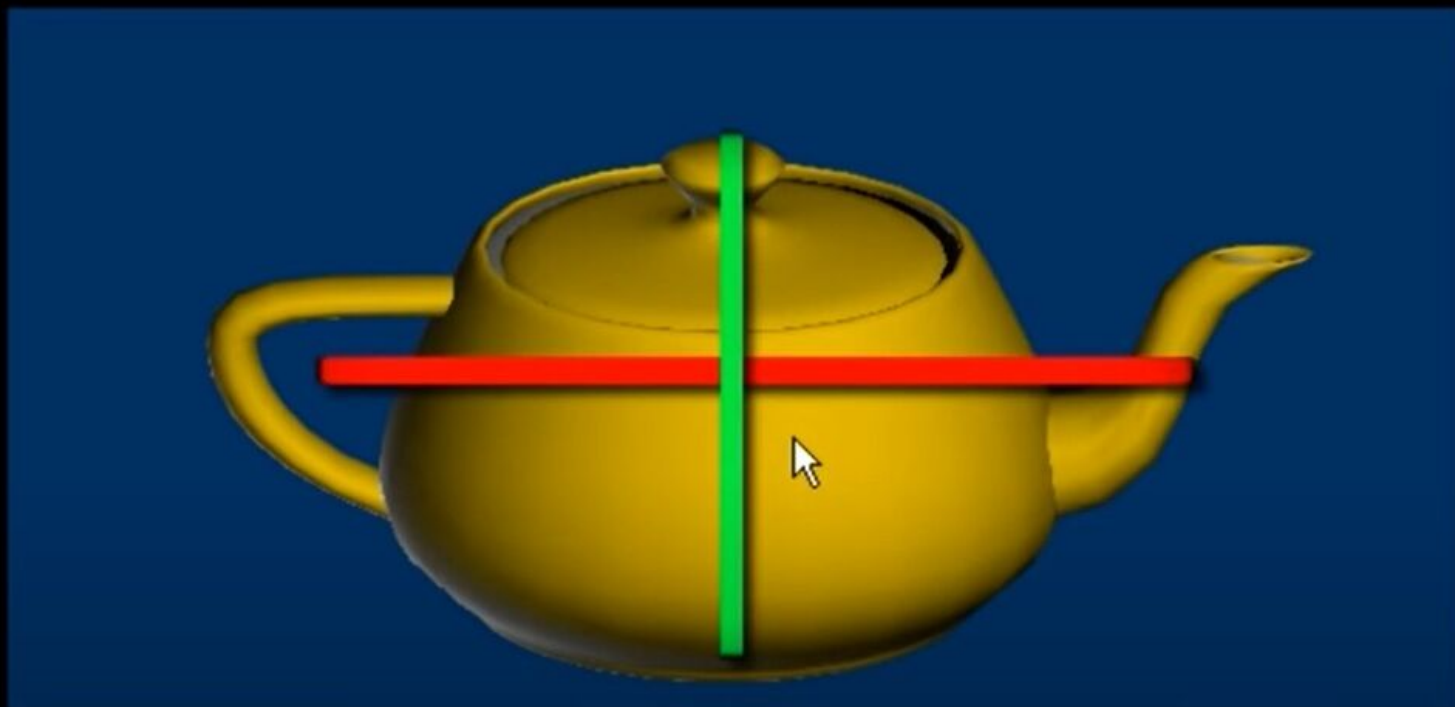




Finding the second longest axis while fixing the first axis...



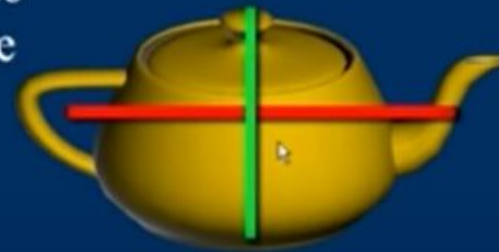




HOW PCA WORKS

Rotate the object around its center to find the best orientation:

- First find the axis so that the object has largest **extend in average** along the axis.
- Rotate the object around the first axis to find the axis that is **perpendicular** to the first axis, and the object has largest extend in average along this axis.



The two axes found are the **first** and the **second** principal component; The **extends in average** along the axes are called the **eigenvalues**.



PCA Algorithm in General

1. Can be applied to high dimensional data.
2. Can be calculated efficiently with computer programs.
3. Use the **covariance matrix** to measure the **average extend** of the data points along any axis.

Popular Applications:

- Visualization of high dimensional data
- Find essential attributes/variables.
- Dimensionality reduction.